

Exponential Functions & Geometric Sequences — Practice Worksheet

Name: _____ Date: _____ Period: _____

Type 1 — Exponential Function from Two Points (MC)

EXAMPLE — Look at this before you begin:

Which exponential function passes through (0, 21) and (1, 3)?

$y = a(b)^x$. At $x=0$: $y=a$, so $a = 21$.

At $x=1$: $y=ab=3$, so $b = 3/21 = 1/7$.

Function: $y = 21(1/7)^x$

Find the exponential function. Circle the correct answer.

Problem 1-A.

Which exponential function passes through the points (0, 18) and (1, 6)?

(A) $y = 6(1/3)^x$

(C) $y = 18(3)^x$

(B) $y = 18(1/3)^x$

(D) $y = 6(3)^x$

Problem 1-B.

Which exponential function passes through the points (0, 8) and (1, 24)?

(A) $y = 24(1/3)^x$

(C) $y = 8(3)^x$

(B) $y = 8(1/3)^x$

(D) $y = 24(3)^x$

Problem 1-C.

Which exponential function passes through the points (0, 50) and (1, 10)?

(A) $y = 50(1/5)^x$

(C) $y = 10(5)^x$

(B) $y = 10(1/5)^x$

(D) $y = 50(5)^x$

Problem 1-D.

Which exponential function passes through the points (0, 4) and (1, 12)?

(A) $y = 12(1/3)^x$

(C) $y = 4(3)^x$

(B) $y = 4(1/3)^x$

(D) $y = 12(3)^x$

Problem 1-E.

Which exponential function passes through the points (0, 100) and (1, 25)?

(A) $y = 100(1/4)^x$

(C) $y = 25(4)^x$

(B) $y = 25(1/4)^x$

(D) $y = 100(4)^x$

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Type 2 — Find the nth Term in a Geometric Sequence (MC)

EXAMPLE — Look at this before you begin:

What is the third term in $f(n) = 2(0.8)^{n-1}$?

Substitute $n=3$: $f(3) = 2(0.8)^2 = 2(0.64) = 1.28$. Answer: 1.28

If given a sequence (no formula), find $r = a_2 / a_1$, then multiply.

Find the requested term. Circle the correct answer.

Problem 2-A.

What is the fourth term in the geometric sequence $f(n) = 5(2)^{n-1}$?

- (A) 20
- (B) 40

- (C) 80
- (D) 10

Problem 2-B.

What is the fifth term in the geometric sequence $f(n) = 3(0.5)^{n-1}$?

- (A) 0.375
- (B) 0.75

- (C) 0.1875
- (D) 1.5

Problem 2-C.

What is the fifth term in the sequence 4, 12, 36, ...?

- (A) 108
- (B) 324

- (C) 972
- (D) 216

Problem 2-D.

What is the sixth term in the sequence 2, 10, 50, ...?

- (A) 1,250
- (B) 6,250

- (C) 31,250
- (D) 250

Problem 2-E.

What is the seventh term in the sequence 64, 32, 16, ...?

- (A) 1
- (B) 2

- (C) 4
- (D) 0.5

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Type 3 — Recursive Rule from Explicit Formula (MC)

EXAMPLE — Look at this before you begin:

Recursive rule for $f(n) = 13(1.3)^{n-1}$?

First term: $f(1) = 13$. Common ratio: 1.3.

Recursive form: $f(1) = 13$, $f(n) = 1.3 \cdot f(n-1)$, $n \geq 2$

Choose the correct recursive rule. Circle the correct answer.

Problem 3-A.

What is the recursive rule for the geometric sequence $f(n) = 13(1.3)^{n-1}$?

- (A) $f(1) = 13$, $f(n-1) = 1.3 \cdot f(n)$, $n \geq 2$
(B) $f(1) = 1.3$, $f(n-1) = 13 \cdot f(n)$, $n \geq 2$

- (C) $f(1) = 13$, $f(n) = 1.3 \cdot f(n-1)$, $n \geq 2$
(D) $f(1) = 1.3$, $f(n) = 13 \cdot f(n-1)$, $n \geq 2$

Problem 3-B.

What is the recursive rule for the geometric sequence $f(n) = 5(2)^{n-1}$?

- (A) $f(1) = 5$, $f(n) = 2 \cdot f(n-1)$, $n \geq 2$
(B) $f(1) = 2$, $f(n) = 5 \cdot f(n-1)$, $n \geq 2$

- (C) $f(1) = 5$, $f(n-1) = 2 \cdot f(n)$, $n \geq 2$
(D) $f(1) = 2$, $f(n-1) = 5 \cdot f(n)$, $n \geq 2$

Problem 3-C.

What is the recursive rule for the geometric sequence $f(n) = 8(0.5)^{n-1}$?

- (A) $f(1) = 0.5$, $f(n) = 8 \cdot f(n-1)$, $n \geq 2$
(B) $f(1) = 8$, $f(n) = 0.5 \cdot f(n-1)$, $n \geq 2$

- (C) $f(1) = 8$, $f(n-1) = 0.5 \cdot f(n)$, $n \geq 2$
(D) $f(1) = 0.5$, $f(n-1) = 8 \cdot f(n)$, $n \geq 2$

Problem 3-D.

What is the recursive rule for the geometric sequence $f(n) = 100(0.1)^{n-1}$?

- (A) $f(1) = 0.1$, $f(n) = 100 \cdot f(n-1)$, $n \geq 2$
(B) $f(1) = 100$, $f(n-1) = 0.1 \cdot f(n)$, $n \geq 2$

- (C) $f(1) = 100$, $f(n) = 0.1 \cdot f(n-1)$, $n \geq 2$
(D) $f(1) = 0.1$, $f(n-1) = 100 \cdot f(n)$, $n \geq 2$

Problem 3-E.

What is the recursive rule for the geometric sequence $f(n) = 7(4)^{n-1}$?

- (A) $f(1) = 4$, $f(n) = 7 \cdot f(n-1)$, $n \geq 2$
(B) $f(1) = 7$, $f(n) = 4 \cdot f(n-1)$, $n \geq 2$

- (C) $f(1) = 7$, $f(n-1) = 4 \cdot f(n)$, $n \geq 2$
(D) $f(1) = 4$, $f(n-1) = 7 \cdot f(n)$, $n \geq 2$

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Type 4 — Linear vs. Exponential Growth (MC)

EXAMPLE — Look at this before you begin:

Linear functions grow by a constant amount each step.

Exponential growth functions multiply by a constant each step.

Exponential growth ALWAYS overtakes linear growth eventually.

Compare the two situations. Circle the correct answer.

Problem 4-A.

Andrew began with \$20 in savings and increased his savings by \$10 every month. Lindsay began with \$5 in savings and doubled her savings every month. After Andrew and Lindsay each saved for 7 months, who saved more money?

- (A) Andrew, because positive linear functions grow faster than exponential growth functions.
- (B) Andrew, because exponential growth functions eventually grow faster than positive linear functions.
- (C) Lindsay, because positive linear functions grow faster than exponential growth functions.
- (D) Lindsay, because exponential growth functions eventually grow faster than positive linear functions.

Problem 4-B.

Marcus began with \$50 in savings and added \$15 every week. Tia began with \$3 in savings and tripled her savings every week. After 6 weeks, who saved more money?

- (A) Marcus, because positive linear functions grow faster than exponential growth functions.
- (B) Marcus, because exponential growth functions eventually grow faster than positive linear functions.
- (C) Tia, because positive linear functions grow faster than exponential growth functions.
- (D) Tia, because exponential growth functions eventually grow faster than positive linear functions.

Problem 4-C.

Two trees were planted on the same day. The pine tree grew 4 inches per year. The maple tree started at 1 inch and doubled in height every year. After 10 years, which tree is taller?

- (A) Pine, because positive linear functions grow faster than exponential growth functions.
- (B) Pine, because exponential growth functions eventually grow faster than positive linear functions.
- (C) Maple, because positive linear functions grow faster than exponential growth functions.
- (D) Maple, because exponential growth functions eventually grow faster than positive linear functions.

Problem 4-D.

A bacteria culture starts with 100 cells and increases by 50 cells per hour. A second culture starts with 10 cells and doubles every hour. After 8 hours, which culture has more cells?

- (A) First, because positive linear functions grow faster than exponential growth functions.
- (B) First, because exponential growth functions eventually grow faster than positive linear functions.
- (C) Second, because positive linear functions grow faster than exponential growth functions.
- (D) Second, because exponential growth functions eventually grow faster than positive linear functions.

Problem 4-E.

Sara started with \$100 in a jar and added \$50 each month. Ben started with \$1 and tripled his savings each month. After 12 months, who saved more?

- (A) Sara, because positive linear functions grow faster than exponential growth functions.
- (B) Sara, because exponential growth functions eventually grow faster than positive linear functions.
- (C) Ben, because positive linear functions grow faster than exponential growth functions.
- (D) Ben, because exponential growth functions eventually grow faster than positive linear functions.

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Type 5 — Recursive Rule from a Table (MC)

EXAMPLE — Look at this before you begin:

Table: $n=1,2,3,4,5$ $f(n)=0.5, 1, 2, 4, 8$

First term: $f(1) = 0.5$. Ratio: $1/0.5 = 2$.

Recursive: $f(1) = 0.5, f(n) = 2 \cdot f(n-1), n \geq 2$

Find the recursive rule. Circle the correct answer.

Problem 5-A.

$n: 1, 2, 3, 4, 5$ $f(n): 0.5, 1, 2, 4, 8$

(A) $f(1) = 2, f(n) = 2 \cdot f(n-1), n \geq 2$

(C) $f(1) = 0.5, f(n) = 2 \cdot f(n-1), n \geq 2$

(B) $f(1) = 2, f(n) = 0.5 \cdot f(n-1), n \geq 2$

(D) $f(1) = 0.5, f(n) = 0.5 \cdot f(n-1), n \geq 2$

Problem 5-B.

$n: 1, 2, 3, 4, 5$ $f(n): 3, 6, 12, 24, 48$

(A) $f(1) = 3, f(n) = 2 \cdot f(n-1), n \geq 2$

(C) $f(1) = 3, f(n) = 3 \cdot f(n-1), n \geq 2$

(B) $f(1) = 2, f(n) = 3 \cdot f(n-1), n \geq 2$

(D) $f(1) = 6, f(n) = 2 \cdot f(n-1), n \geq 2$

Problem 5-C.

$n: 1, 2, 3, 4, 5$ $f(n): 100, 50, 25, 12.5, 6.25$

(A) $f(1) = 100, f(n) = 2 \cdot f(n-1), n \geq 2$

(C) $f(1) = 0.5, f(n) = 100 \cdot f(n-1), n \geq 2$

(B) $f(1) = 100, f(n) = 0.5 \cdot f(n-1), n \geq 2$

(D) $f(1) = 50, f(n) = 0.5 \cdot f(n-1), n \geq 2$

Problem 5-D.

$n: 1, 2, 3, 4, 5$ $f(n): 4, 12, 36, 108, 324$

(A) $f(1) = 3, f(n) = 4 \cdot f(n-1), n \geq 2$

(C) $f(1) = 4, f(n) = 3 \cdot f(n-1), n \geq 2$

(B) $f(1) = 4, f(n) = 4 \cdot f(n-1), n \geq 2$

(D) $f(1) = 12, f(n) = 3 \cdot f(n-1), n \geq 2$

Problem 5-E.

$n: 1, 2, 3, 4, 5$ $f(n): 5, 1, 0.2, 0.04, 0.008$

(A) $f(1) = 5, f(n) = 0.2 \cdot f(n-1), n \geq 2$

(C) $f(1) = 5, f(n) = 5 \cdot f(n-1), n \geq 2$

(B) $f(1) = 0.2, f(n) = 5 \cdot f(n-1), n \geq 2$

(D) $f(1) = 1, f(n) = 0.2 \cdot f(n-1), n \geq 2$

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Type 6 — Classify: Arithmetic or Geometric

EXAMPLE — Look at this before you begin:

Sequence: 5, 9, 13, 17, 21, ... Differences: +4 each time -> Arithmetic.

Sequence: 3, 6, 12, 24, 48, ... Ratios: x2 each time -> Geometric.

Arithmetic = constant DIFFERENCE. Geometric = constant RATIO.

Place an X in the table to show whether each sequence is arithmetic or geometric.

	Arithmetic	Geometric
78, 69, 60, 51, 42, ...		
4, 12, 36, 108, 324, ...		
-0.6, 0.2, 1, 1.8, 2.6, ...		
225, -22.5, 2.25, -0.225, 0.0225, ...		
100, 50, 25, 12.5, 6.25, ...		

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Type 7 — Common Ratio (Short Answer)

EXAMPLE — Look at this before you begin:

Find the common ratio of 6, 18, 54, 162, ...

$r = (\text{any term}) / (\text{previous term})$. $18/6 = 3$. $54/18 = 3$. Confirmed.

Common ratio = 3.

Find the common ratio of each sequence. Show your work.

Problem 7-A.

$r =$ _____

What is the common ratio of the sequence 81, 27, 9, 3, 1, ...?

Problem 7-B.

$r =$ _____

What is the common ratio of the sequence 2, 6, 18, 54, ...?

Problem 7-C.

$r =$ _____

What is the common ratio of the sequence 1000, 200, 40, 8, ...?

Problem 7-D.

$r =$ _____

What is the common ratio of the sequence 5, -10, 20, -40, ...?

Problem 7-E.

$r =$ _____

What is the common ratio of the sequence 128, 64, 32, 16, ...?

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Type 8 — Evaluate Exponential Function (Short Answer)

EXAMPLE — Look at this before you begin:

If $f(x) = 256(0.5)^x$, what is $f(4)$?

Substitute $x=4$: $f(4) = 256(0.5)^4 = 256(0.0625) = 16$.

Answer: $f(4) = 16$

Evaluate each function at the given value. Show your work.

Problem 8-A.

If $f(x) = 625(0.8)^x$, what is the value of $f(4)$?

Problem 8-B.

If $f(x) = 3(2)^x$, what is the value of $f(5)$?

Problem 8-C.

If $f(x) = 100(0.5)^x$, what is the value of $f(3)$?

Problem 8-D.

If $f(x) = 4(3)^x$, what is the value of $f(4)$?

Problem 8-E.

If $f(x) = 1024(0.5)^x$, what is the value of $f(6)$?

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Type 9 — Write the Explicit Rule (Short Answer)

EXAMPLE — Look at this before you begin:

A geometric sequence has first term 23 and common ratio 7.5.

Explicit rule format: $f(n) = a \cdot r^{n-1}$

Substitute: $f(n) = 23(7.5)^{n-1}$

Write the explicit rule for each geometric sequence in the form $f(n) = a(r)^{n-1}$.

Problem 9-A.

A geometric sequence has a first term of 6 and a common ratio of 4. What is the explicit rule for this sequence?

$f(n) =$ _____

Problem 9-B.

A geometric sequence has a first term of 100 and a common ratio of 0.5. What is the explicit rule for this sequence?

$f(n) =$ _____

Problem 9-C.

A geometric sequence has a first term of 2 and a common ratio of 5. What is the explicit rule for this sequence?

$f(n) =$ _____

Problem 9-D.

A geometric sequence has a first term of 81 and a common ratio of $\frac{1}{3}$. What is the explicit rule for this sequence?

$f(n) =$ _____

Problem 9-E.

A geometric sequence has a first term of 7 and a common ratio of 2. What is the explicit rule for this sequence?

$f(n) =$ _____

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Answer Key

Type 1 — Exponential Function from Two Points

- 1-A (B) $y = 18(1/3)^x$
- 1-B (C) $y = 8(3)^x$
- 1-C (A) $y = 50(1/5)^x$
- 1-D (C) $y = 4(3)^x$
- 1-E (A) $y = 100(1/4)^x$

Type 2 — nth Term in a Geometric Sequence

- 2-A (B) 40
- 2-B (C) 0.1875
- 2-C (B) 324
- 2-D (B) 6,250
- 2-E (A) 1

Type 3 — Recursive Rule from Explicit Formula

- 3-A (C) $f(1) = 13$, $f(n) = 1.3 \cdot f(n-1)$
- 3-B (A) $f(1) = 5$, $f(n) = 2 \cdot f(n-1)$
- 3-C (B) $f(1) = 8$, $f(n) = 0.5 \cdot f(n-1)$
- 3-D (C) $f(1) = 100$, $f(n) = 0.1 \cdot f(n-1)$
- 3-E (B) $f(1) = 7$, $f(n) = 4 \cdot f(n-1)$

Type 4 — Linear vs. Exponential Growth

- 4-A (D) Lindsay (exponential eventually grows faster)
- 4-B (D) Tia (exponential eventually grows faster)
- 4-C (D) Maple (exponential eventually grows faster)
- 4-D (D) Second culture (exponential eventually grows faster)
- 4-E (D) Ben (exponential eventually grows faster)

Type 5 — Recursive Rule from a Table

- 5-A (C) $f(1) = 0.5$, $f(n) = 2 \cdot f(n-1)$
- 5-B (A) $f(1) = 3$, $f(n) = 2 \cdot f(n-1)$
- 5-C (B) $f(1) = 100$, $f(n) = 0.5 \cdot f(n-1)$
- 5-D (C) $f(1) = 4$, $f(n) = 3 \cdot f(n-1)$
- 5-E (A) $f(1) = 5$, $f(n) = 0.2 \cdot f(n-1)$

Type 6 — Arithmetic vs. Geometric

- 78, 69, 60, 51, 42, ... -> Arithmetic ($d = -9$)
- 4, 12, 36, 108, 324, ... -> Geometric ($r = 3$)
- 0.6, 0.2, 1, 1.8, 2.6, ... -> Arithmetic ($d = +0.8$)
- 225, -22.5, 2.25, -0.225, 0.0225, ... -> Geometric ($r = -0.1$)
- 100, 50, 25, 12.5, 6.25, ... -> Geometric ($r = 0.5$)

Type 7 — Common Ratio

- 7-A $r = 1/3$
- 7-B $r = 3$
- 7-C $r = 1/5$
- 7-D $r = -2$
- 7-E $r = 1/2$

Type 8 — Evaluate Exponential Function

- 8-A $f(4) = 256$
- 8-B $f(5) = 96$
- 8-C $f(3) = 12.5$
- 8-D $f(4) = 324$
- 8-E $f(6) = 16$

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Answer Key (continued)

Type 9 — Explicit Rule

9-A $f(n) = 6(4)^{n-1}$

9-B $f(n) = 100(0.5)^{n-1}$

9-C $f(n) = 2(5)^{n-1}$

9-D $f(n) = 81(1/3)^{n-1}$

9-E $f(n) = 7(2)^{n-1}$